

Artificial Intelligence (CS-203)

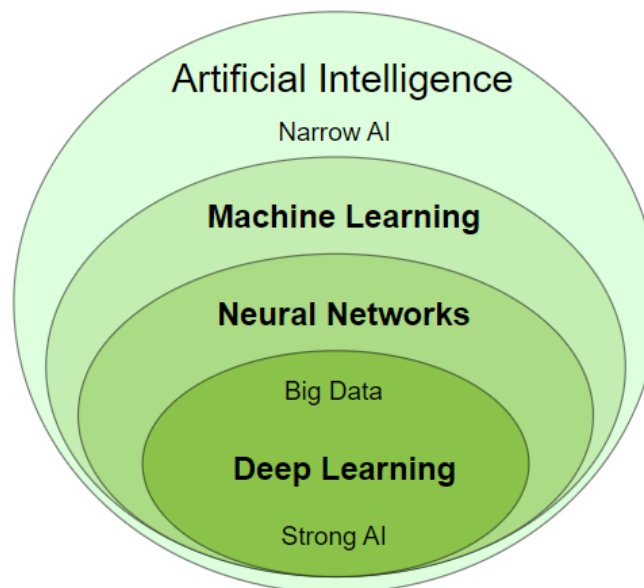
Objectives:

- What is Machine learning?
 - Types of machine learning?
 - Challenges with machine learning
 - Terms used in machine learning
 - Steps involved in training Machine learning Model
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Machine Learning

Machine Learning is the science (and art) of programming computers so they can learn from data. Here is a slightly more general definition:

“Machine Learning is the field of study that gives computers the ability to learn without being explicitly programmed.” (Arthur Samuel, 1959)



Why use Machine learning?

Rule base systems, GPS, ELIZA any many others following the traditional approach explained in this section. The traditional approach being used in all classical AI applications is given below in the figure.

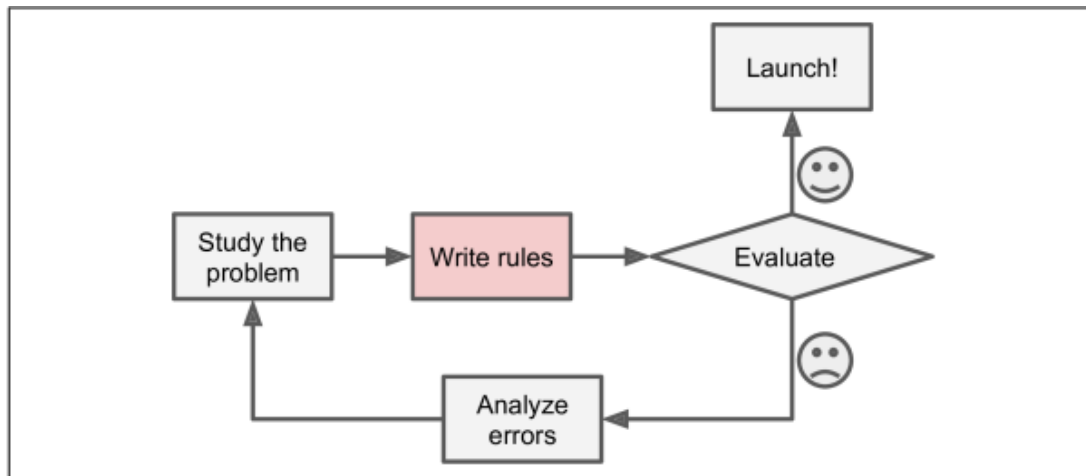


Figure 1-1. The traditional approach

The next figure explains the strategy being followed by machine learning algorithms which clearly shows the difference between both of the approaches.

Machine Learning is great for:

- Problems for which existing solutions require a lot of hand-tuning or long lists of rules: one Machine Learning algorithm can often simplify code and perform better.
- Complex problems for which there is no good solution at all using a traditional approach: the best Machine Learning techniques can find a solution.
- Fluctuating environments: a Machine Learning system can adapt to new data.
- Getting insights about complex problems and large amounts of data.

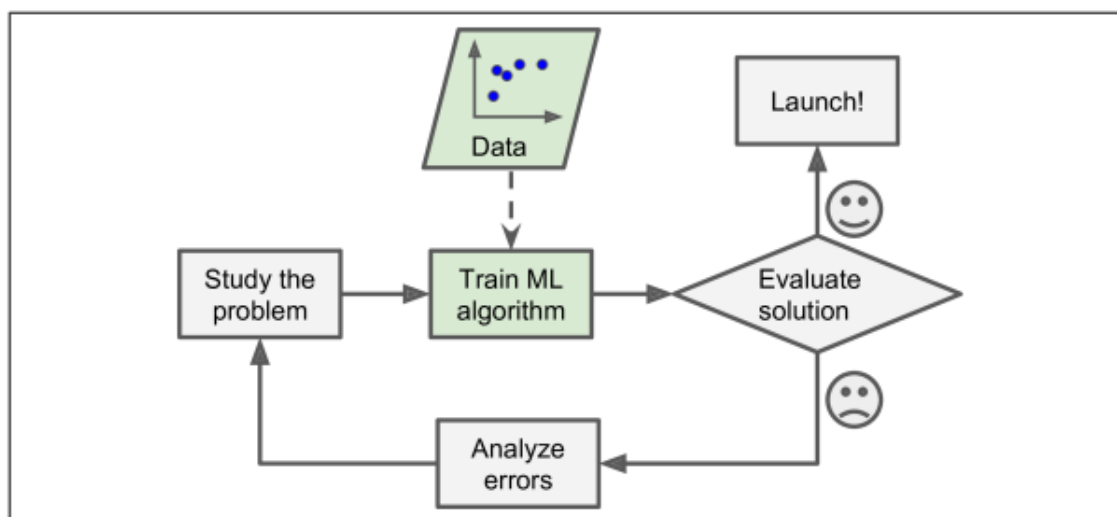


Figure 1-2. Machine Learning approach

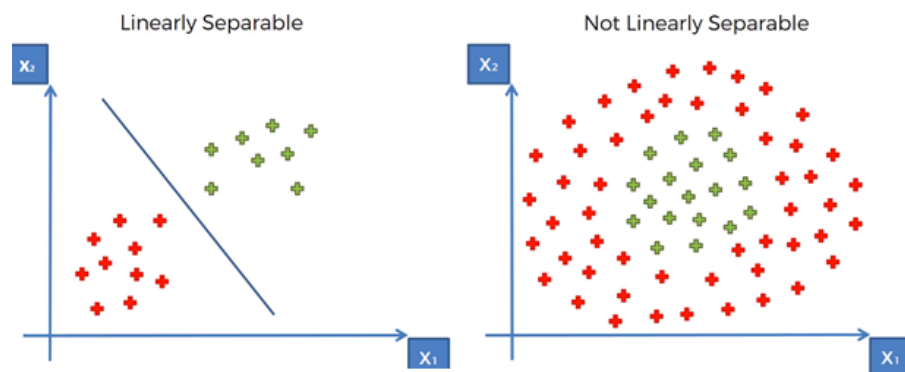
Types of Machine Learning

Supervised Learning

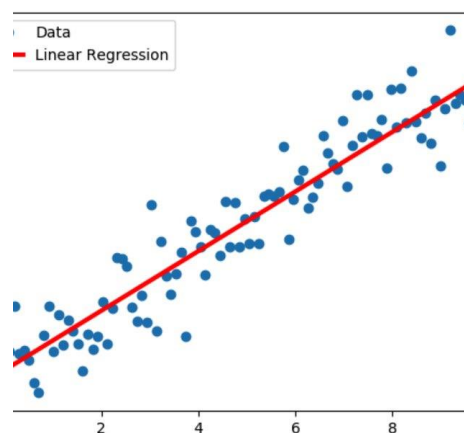
Supervised learning uses a training set to teach models to yield the desired output. This training dataset includes inputs and correct outputs, which allow the model to learn over time. The algorithm measures its accuracy through the loss function, adjusting until the error has been sufficiently minimized.

Supervised learning can be separated into two types of problems: classification and regression.

Classification uses an algorithm to accurately assign test data into specific categories. It recognizes specific entities within the dataset and attempts to draw some conclusions on how those entities should be labeled or defined. Common classification algorithms are linear classifiers, support vector machines (SVM), decision trees, k-nearest neighbor, and random forest, which are described in more detail below.

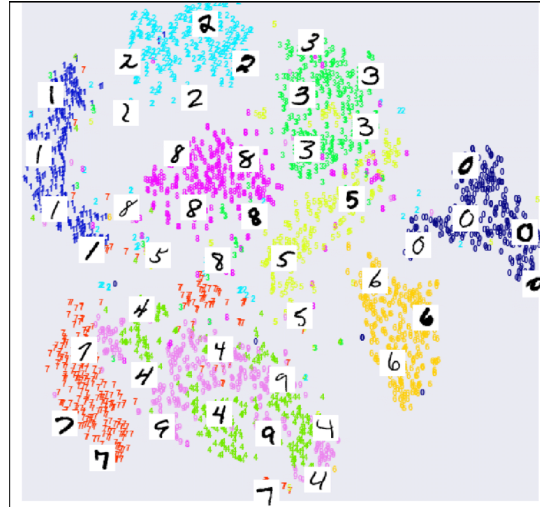


Regression is used to understand the relationship between dependent and independent variables. It is commonly used to make projections, such as for sales revenue for a given business. Linear regression, logistical regression, and polynomial regression are popular regression algorithms.



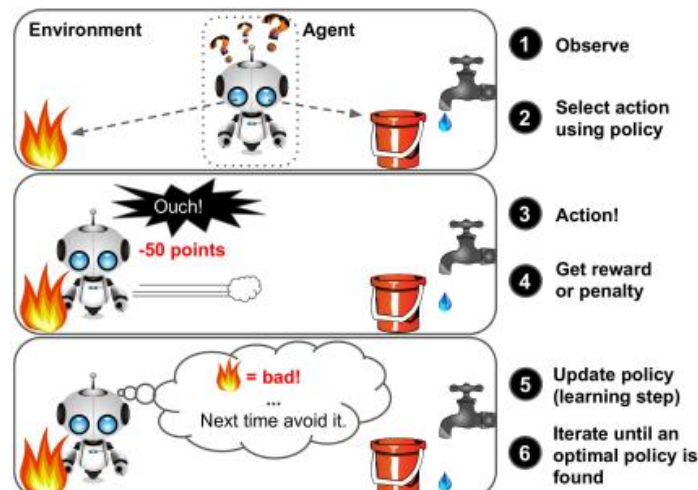
Unsupervised Learning

Unlike supervised learning, unsupervised learning uses unlabeled data. From that data, it discovers patterns that help solve for **clustering** or association problems. This is particularly useful when subject matter experts are unsure of common properties within a data set. Common clustering algorithms are hierarchical, k-means, and Gaussian mixture models.



Reinforcement Learning

Reinforcement Learning is a very different approach. The learning system, called an agent in this context, can observe the environment, select and perform actions, and get rewards in return (or penalties in the form of negative rewards). It must then learn by itself what is the best strategy, called a policy, to get the most reward over time. A policy defines what action the agent should choose when it is in a given situation.



Differences among Supervised, Unsupervised and Reinforcement Learning

Supervised Learning	Unsupervised Learning	Reinforcement Learning
Data provided is labeled data, with output values specified	Data provided is unlabeled data, the outputs are not specified, machine makes its own prediction	The machine learns from its environment using rewards and errors
Used to solve Regression and classification problems	Used to solve Association and clustering problems	Used to solve Reward based problems
Labeled data is used	Unlabeled data is used	No predefined data is used
External Supervision	No supervision	No supervision
Solves problems by mapping labeled input to known output	Solves problems by understanding patterns and discovering output	Follows Trail and Error problem solving approach

Main Challenges of Machine Learning

- Insufficient Quantity of Training Data
- Nonrepresentative Training Data
- Poor-Quality Data
- Irrelevant Features
- Overfitting the Training Data
- Underfitting the Training Data

Terms used in machine learning

Data: Any material which is being used for learning purpose is data, it can be in the form of images, audios, videos, text etc.

Sample: The data we use for representing one example is called a sample.

Features: The characteristics of one object which makes it different from other objects are called features. (features should be in numeric form)

Training/Testing data: To make a machine do learn some data we divide the dataset into two parts, training dataset (to make machine learn data) and testing dataset (to check/test whether machine learnt something or not).

Performance Evaluation: Testing the performance of a model whether it had learnt something or not for which test data set is used. The samples in test dataset are unknown for model as it was not trained on them in this way we can see how well it perform on unseen samples.

Performance Metrics: An amount which show how good our trained model is on test dataset. Accuracy is a performance metric which tell the performance of a model.

$$Acc = \frac{\text{Correct predictions}}{\text{total test samples}} * 100$$

Label/Class: Type of samples present in a dataset e.g Trucks and cars, nine digits of numbers etc.

Predicted label: The class a ML model is predicting after getting trained (represented as $\hat{y}$).

Actual Label: the actual class a sample belongs to (represented as y).

Steps involved in training Machine learning Model

